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Adjustment Formula for Harmonizing Triglyceride Values in the Korea National Health and Nutrition Examination Survey, 2005–2022

Rihwa Choi, M.D., Ph.D.[1,2,*], Jong Do Seo, M.D.[3,*], Eun-Jung Cho, M.D., Ph.D.[4], Woochang Lee, M.D., Ph.D.[5], and Yeo-Min Yun, M.D., Ph.D.[3,6]

[1]Department of Laboratory Medicine, Green Cross Laboratories (GC Labs), Yongin, Korea; [2]Department of Laboratory Medicine and Genetics, Sungkyunkwan University School of Medicine, Seoul, Korea; [3]Department of Laboratory Medicine, Konkuk University Medical Center, Seoul, Korea; [4]Department of Laboratory Medicine, Hallym University Dongtan Sacred Heart Hospital, Hallym University College of Medicine, Hwaseong, Korea; [5]Department of Laboratory Medicine, Asan Medical Center, University of Ulsan College of Medicine, Seoul, Korea; [6]Department of Laboratory Medicine, Konkuk University School of Medicine, Seoul, Korea

Background: Korea National Health and Nutrition Examination Survey (KNHANES) triglyceride testing changed from the glycerol blanking method (2005–2021) to the glycerol non-blanking method (2022). We converted triglyceride data from 2005–2021 to that obtained since 2022 with different analytical methods.

Methods: To develop a conversion equation, 98 fresh serum specimen pairs were compared using Passing–Bablok regression analysis. Implications of the conversion equation on epidemiological data were evaluated using KNHANES data from 2019–2021. Bias estimations determined using the Lipid Standardization Program (LSP) of the United States Centers for Disease Control and Prevention (CDC) enhanced the accuracy and comparability of the triglyceride results.

Results: Triglyceride concentrations measured via the glycerol non-blanking method were 10.7 mg/dL (0.12 mmol/L, 10.0%) higher than those from the glycerol blanking method, with a 9.9 mg/dL (0.11 mmol/L, 5.0%) difference at a concentration of 200 mg/dL (2.26 mmol/L, N=98). The conversion equation y (glycerol non-blanking, 2022)=11.94+0.99x (glycerol blanking, 2005–2021) changed the mean triglyceride concentrations of the KNHANES 2019–2021 data (N=16,015) from 123.7 mg/dL (1.40 mmol/L, 95% confidence interval [CI]: 122.2–125.1 mg/dL [1.38–1.41 mmol/L]) to 134.3 mg/dL (1.52 mmol/L, 95% CI: 132.9–135.8 mg/dL [1.50–1.53 mmol/L]). Since 2022, bias monitoring using the CDC's LSP has remained within a 5.0% limit.

Conclusions: KNHANES triglyceride values in 2022 (non-blanking) were substantially higher than those from 2005–2021 (blanking). Conversion equations helped effectively adjust 2005–2021 data. Researchers should consider adjusting the KNHANES triglyceride data based on their study characteristics.

Key Words: Calibration, Data Processing, Korea National Health and Nutrition Examination Survey, Standardization, Triglycerides

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Corresponding author:
Yeo-Min Yun, M.D., Ph.D.
Department of Laboratory Medicine,
Konkuk University School of Medicine,
120-1 Neungdong-ro, Gwangjin-gu,
Seoul 05030, Korea
E-mail: ymyun@kuh.ac.kr

*These authors contributed equally to this study as co-first authors.

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INTRODUCTION

Triglycerides, the primary constituents of body fat in humans, are crucial biomarkers for cardiovascular health and metabolic disorders [1, 2]. Accurately assessing serum triglyceride concentrations is essential for epidemiological studies, public health monitoring, and individual patient management [3]. Interpreting triglyceride data requires the consideration of various factors to ensure accuracy and relevance [2].

Total glyceride concentrations constitute the sum of monoglyceride, diglyceride, triglyceride, and free glycerol concentrations [4–9]. For triglyceride measurements, the entity measured is glycerol, which is released following the hydrolysis of tri-, di-, and monoglycerides. The presence of free glycerol is considered an interference [7–10]. In clinical practice, confusion often arises regarding the use of triglycerides versus total glycerides as clinical biomarkers [7, 10]. Analytically, the glycerol non-blanking method, which measures total glycerides, has been used most commonly [7, 10]. In contrast, the glycerol blanking method, which eliminates free glycerol, results in lower triglyceride concentrations than with the glycerol non-blanking method. This phenomenon has also been observed in external quality-assurance programs, including the United States Centers for Disease Control and Prevention (CDC) Lipid Standardization Program (LSP) and the Korean Association of External Quality Assessment Service (KAEQAS) [7–9]. Given that the analytical method for measuring triglycerides in the Korea National Health and Nutrition Examination Survey (KNHANES) changed from glycerol blanking (2005–2021) to glycerol non-blanking (2022), a conversion equation to adjust the differences in triglyceride concentrations is necessary.

In Korea, KNHANES has been instrumental in understanding population health trends and shaping public health policy [11, 12]. Since 1998, KNHANES has monitored triglyceride concentrations, utilizing various analytical methods that reflect technological advances and shifts in scientific understanding [5, 6, 12, 13]. The glycerol blanking method, employed in the KNHANES from 2005 to 2021, offers the advantage of reduced interference from free glycerol. However, comparing its calibrator traceability with the CDC LSP, which assigns target values using methods without glycerol blanking, reveals substantial differences [3, 6, 7, 14, 15]. This shift in analytical methods mirrors broader international trends toward standardizing and enhancing the accuracy and comparability of lipid measurements [6, 7, 14]. Understanding the implications of methodological changes is crucial for interpreting historical data and ensuring continuity in longitudinal studies within KNHANES, which is representative of the Korean population [4, 5].

We developed a conversion equation to harmonize triglyceride 2005–2021 KNHANES data (glycerol blanking methods) with 2022 KNHANES data (glycerol non-blanking methods), enabling adjustment of the 2005–2021 data and illustrating the potential impacts of these methodological changes on research data. The need for such analyses is emphasized by the growing global focus on comparable and reliable health data, which is vital for international health comparisons and monitoring population health trends over time [15, 16].

METHODS

Differences in the analytical methods used in the KNHANES from 2005 to 2022 are summarized in Table 1. All procedures and protocols of the KNHANES were approved by the Institutional Review Board of the Korea Disease Control and Prevention Agency (KDCA; approval number 2018-01-03-4C-A), with a waiver of informed consent granted under this approval. This study was conducted using anonymized data for the research task named "Quality Control of the Clinical Laboratory for the Korea National Health and Nutrition Examination Survey (KNHANES)," which covers the 2019–2021 (8th) and 2022–2024 (9th) phases. The task, abbreviated as the "Quality Control for KNHANES" task, was designed to improve the quality and reliability of the KNHANES and was conducted under the oversight of the KDCA. The findings are publicly accessible through the Policy Research Information Service & Management database [17]. To develop a conversion equation for triglyceride measurements, we utilized data from a method-comparison study (Quality Control for KNHANES). The comparison study involved comparing data from Seegene Medical Foundation (Neodin), which performed KNHANES triglyceride tests using the glycerol blanking method from 2008 to 2021, and GC Labs, which has been conducting triglyceride tests using the glycerol non-blanking method since 2022.

This study involved 98 fresh leftover serum samples from 98 individuals undergoing routine clinical diagnosis for serum triglycerides at an outpatient clinic or during a healthcare checkup. On January 6, 2022, the specimens were collected at the Asan Medical Center Laboratory, the affiliated institution of the principal investigator for the Quality Control for KNHANES task. Specimens were collected to cover the analytical measurement interval shared by both methods (9–885 mg/dL, 0.10–10.0 mmol/L) and included specimens distributed across low-to-high

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Table 1. Analytical methods used for triglyceride measurement in the KNHANES

Year	2005, 2007, and 2008*	2008–2017†	2018–2021	Since 2022
Measurement principle	Enzymatic, with glycerol blanking	Enzymatic, with glycerol blanking	Enzymatic, with glycerol blanking	Direct enzymatic, without glycerol blanking
Reagent kit	TRIG (Siemens‡, USA)	Pureato S TG-N (Sekisui§, Japan)	Qualigent TG (Sekisui)	TRIGL (Roche, China)
Calibrator name	Not available	Cholestest N calibrator (Sekisui)	Qualigent N calibrator (Sekisui)	Calibrator for automated systems (Roche, Germany)
Calibrator traceability	Not available	Reference material produced by HECTEF (Japan)	Reference material produced by HECTEF (Japan)	IDMS
Analyzer	Advia 1650 (Siemens, USA)	Hitachi 7600 (Hitachi, Japan)	Labospect 008AS (Hitachi, Japan)	Cobas 8000 c702 (Roche, Germany)
Laboratory	Seoul Clinical Laboratories	Seegene Medical Foundation (Neodin)‖	Seegene Medical Foundation	GC Labs
US FDA clearance	US FDA cleared	Non-US FDA cleared	Non-US FDA cleared	US FDA cleared
Free glycerol treatment	Glycerol blanking (free glycerol eliminated)	Glycerol blanking (free glycerol eliminated)	Glycerol blanking (free glycerol eliminated)	Glycerol non-blanking (free glycerol included)
Characteristics of analytical methods	Calibrator traceability required conversion for data comparisons with the CDC LSP due to negative bias			Calibrator traceability is assured and comparable with international data by the CDC LSP in accordance with NCEP guidelines
Conversion equation	KNHANES 2022 (glycerol non-blanking) = 11.94+0.99 × KNHANES 2005–2021 (glycerol blanking)			

*Triglycerides were measured at Seoul Clinical Laboratories from July 1, 2007 to February 15, 2008.
†Triglycerides were measured at Seegene Medical Foundation (Neodin) from February 20, 2008 to December 31, 2021.
‡The manufacturer Bayer (used in 2005) merged with Siemens in 2007.
§The manufacturer Daiichi (used in 2008) merged with Sekisui in 2009.
‖The laboratory changed its name from Neodin (2008–2013) to Seegene Medical Foundation in 2014.
According to the NCEP guidelines, analytical methods traceable to the CDC LSP are recommended. During the study period, all reagents and calibrator lots were evaluated by the KDCA. The study, titled 'Quality Control of the Clinical Laboratory for the Korea National Health and Nutrition Examination Survey (KNHANES),' was supported and monitored by the KDCA.
Abbreviations: CDC, Centers for Disease Control and Prevention in the US; FDA, Food and Drug Administration; HECTEF: Health Care Technology Foundation; IDMS, isotope dilution mass spectrometry; KDCA, Korea Disease Control and Prevention Agency; KNHANES, Korea National Health and Nutrition Examination Surveys; LSP, Lipids Standardization Program; NCEP, National Cholesterol Education Program.

concentrations according to the National Cholesterol Education Program (NCEP) Adult Treatment Panel (ATP) III criteria (normal concentrations, <150 mg/dL [1.70 mmol/L]; very high concentrations, ≥500 mg/dL [5.65 mmol/L]) [1]. The specimens were prepared and refrigerated at the Asan Medical Center Laboratory on the same day of collection. Each sample was divided into two aliquots to provide identical testing materials for comparative analysis. The aliquots were concurrently sent to Seegene Medical Foundation and GC Labs (1 h transport time) on the specimen collection date under refrigeration, where the triglycerides were analyzed immediately. At Seegene Medical Foundation, triglyceride analysis was conducted using the glycerol blanking method with Qualigent TG Reagent Kits (Sekisui, Japan; lot 148RBS) and a Qualigent N calibrator (Sekisui; lot 119RES) on a Labospect 008AS analyzer (Hitachi, Japan), and the method was designated as the X method. Conversely, GC Labs performed measurements using a non-glycerol blanking method with TRIGL Reagent Kits (Roche, China; lot 599110) and a certified functional activity standard calibrator (Roche, Germany; lot 539840) on a Cobas 8000 c702 analyzer (Roche, Germany), where this method was designated as the Y method. Both laboratories analyzed the triglycerides in duplicate to ensure the reliability and reproducibility of the data.

Statistical analysis and derivation of the conversion equation

Triglyceride values from duplicate measurements at each laboratory, along with the mean of these duplicates, were used for comparative analysis. The entire sample preparation, analysis, and statistical evaluation rigorously adhered to CLSI guidelines (EP09c–ED3:2018) to ensure that the study conformed to internationally recognized laboratory practices for method comparisons [18]. Passing–Bablok regression analysis was employed to derive the conversion equation between both methods. This non-parametric statistical method is particularly well suited for method-comparison studies as it does not assume a normal distribution of residuals and is robust against outliers [5, 18].

Evaluating the potential impacts of the conversion equation

KNHANES data from 1998–2022 are publicly accessible [12]. KNHANES triglyceride data from 2019–2021 (with glycerol blanking; 8th phase) and 2022 (without glycerol blanking; 9th phase) were compared to illustrate the potential impacts of the conversion equation, as an example. Quantitative data were analyzed using a mountain plot. Because triglyceride concentrations fluctuate owing to recent dietary intake and pregnancy status, triglyceride data from pregnant women and data with fasting times of <12 hrs were excluded from our analysis [19]. Triglyceride data for all age groups (≥10 yrs) were included. To compare triglyceride data before and after conversion, a kernel density estimation was used to delineate the shape of the distribution (revealing clustering, skewness, or bimodality) and to compare continuous variable distributions across different groups [4, 20]. According to the NCEP ATP III criteria, ≥200 mg/dL (2.26 mmol/L) was regarded as hypertriglyceridemia [1, 11, 19]. An unweighted prevalence estimation was performed for our example analysis.

Estimating bias with the CDC LSP

The accuracies of the analytical methods used for triglyceride measurements were assured through participation in the CDC LSP, which involves frozen serum pools. This assurance covers the KNHANES 2018–2022 period for which results are available. The CDC LSP involves unaltered pooled serum samples (which are highly similar to individual patient samples) as blind reference samples to evaluate the analytical performance of laboratory tests [6, 7, 14, 15]. The commutability and specimen stability of each serum sample were established and assured by the CDC [1, 6, 7, 14]. Participants received frozen sample sets and analyzed one sample weekly after thawing for 12 weeks, according to a preset schedule [6, 7, 14, 15]. The Seegene Medical Foundation participated in the CDC LSP from 2018 to 2021. Starting in 2022, GC Labs took over participation. During their respective participation periods, each laboratory provided results to the KDCA as part of the Quality Control for KNHANES task. The bias estimation derived from the CDC LSP target values was analyzed to compare triglyceride data obtained using either the glycerol blanking or non-blanking method, serving as another example of data comparison [6, 7, 14, 15].

RESULTS

Fig. 1 presents the outline of our study. Data from 98 fresh, refrigerated serum specimens demonstrated significantly higher triglyceride concentrations with the glycerol non-blanking method (2022) than with the glycerol blanking method (2005–2021), with an average difference of 10.7 mg/dL (0.12 mmol/L, 95% CI: 8.9–12.6 mg/dL [0.10–0.14 mmol/L]) and an average percent difference of 10.0% (95% CI: 8.3–11.7). At the 200 mg/dL (2.26 mmol/L) triglyceride concentration (the cutoff for hypertriglyceridemia according to NCEP ATP III criteria and the KNHANES annual prevalence-assessment standards), the estimated and percent differences were 9.9 mg/dL (0.11 mmol/L) and 5.0%, respectively. The estimated differences at the cutoffs for normal

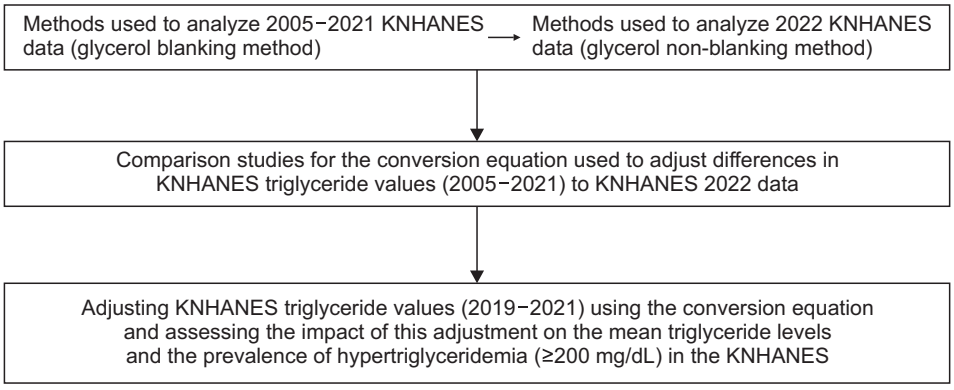


Fig. 1. Flow diagram for the study. As the analytical methods for triglyceride measurement in KNHANES transitioned from glycerol blanking methods (2005–2021) to glycerol non-blanking methods (2022 onward), a conversion equation was developed to adjust for differences in triglyceride values. The equation was derived from comparison studies that evaluated triglyceride values obtained using glycerol blanking methods (2021) and glycerol non-blanking methods (2022) at the KNHANES laboratory. The application of this conversion equation was assessed using KNHANES triglyceride values from 2019 to 2021 as an example. A triglyceride concentration of 200 mg/dL is equivalent to 2.26 mmol/L.
Abbreviation: KNHANES, Korea National Health and Nutrition Examination Survey.

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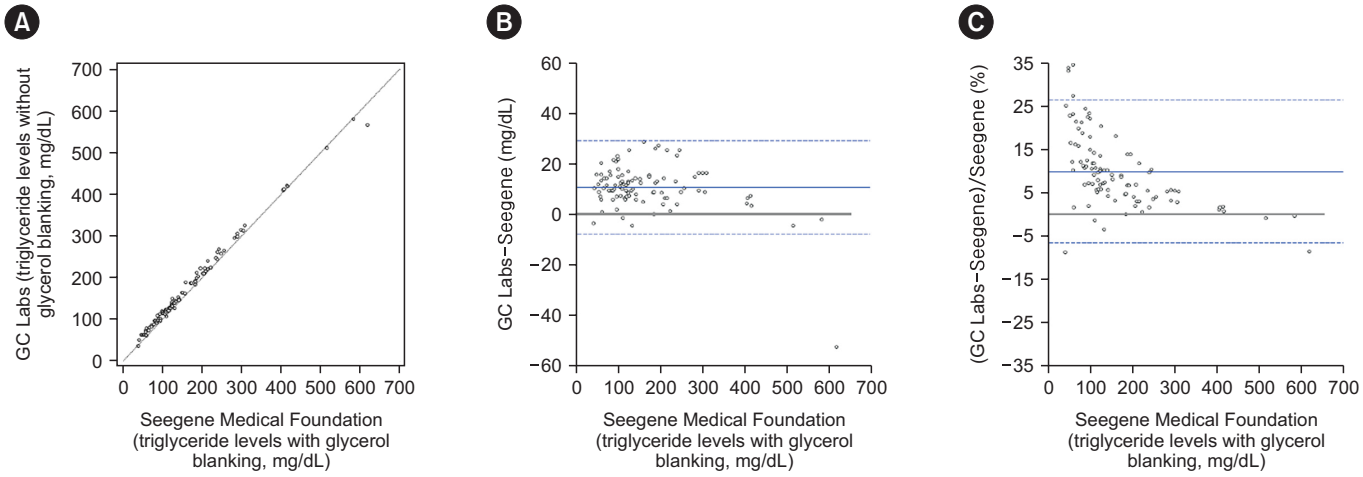


Fig. 2. Comparison study of triglyceride measurements. (A) Scatter plot with Passing–Bablok regression analysis depicting the relationship between triglyceride measurements obtained using the X method with KNHANES 2021 data (glycerol blanking) at Seegene Medical Foundation and the Y method with KNHANES 2022 data (glycerol non-blanking) at GC Labs. The solid 45° line indicates the line of identity, and points represent individual paired measurements from the two methods. (B) Bland–Altman plot showing differences in triglyceride concentrations (mg/dL). (C) Bland–Altman plot showing percent differences in triglyceride concentrations (%). The solid blue line in the Bland–Altman plots represents the mean difference and mean percent difference, with a 95% confidence interval indicated by dotted blue lines. Solid gray lines are drawn at zero.
Abbreviation: KNHANES, Korea National Health and Nutrition Examination Survey.

(150 mg/dL [1.70 mmol/L]) and very high (500 mg/dL [5.65 mmol/L]) concentrations were 10.4 mg/dL (0.12 mmol/L, 6.9%) and 6.9 mg/dL (0.08 mmol/L, 1.4%), respectively.

Passing–Bablok regression analysis indicated a high correlation between both methods with a correlation coefficient (r=0.997). This regression analysis established the relationship between both methods with the equation y (mg/dL)=11.94 (mg/dL)+0.99x, where x represents triglyceride measurements using the glycerol blanking method at Seegene Medical Foundation, and y denotes corresponding values from the non-glycerol blanking method at GC Labs (Fig. 2). The 95% CI of the slope includes 1.0 (0.9772–1.013), but the 95% CI of the intercept does not include 0.0 (9.50–14.76), signifying significant differences between the analytical methods.

The conversion equation y (mg/dL; glycerol non-blanking, 2022)=11.94 (mg/dL)+0.99x (glycerol blanking, 2005–2021) was applied to 2019–2021 data from the KNHANES (N=16,015) to demonstrate data conversion. The mean triglyceride concentrations measured in the KNHANES (2019–2021; N=16,015) shifted from 123.7 mg/dL (1.40 mmol/L, 95% CI: 122.2–125.1 mg/dL [1.38–1.41 mmol/L]) to 134.3 mg/dL (1.52 mmol/L, 95% CI: 132.9–135.8 mg/dL [1.50–1.53 mmol/L]) after conversion. Fig. 3 displays Kernel density plots of serum triglyceride concentrations from the KNHANES. Before conversion, the KNHANES data from 2019–2021 and 2022 (blue and green lines, respectively) showed similar distributions. After applying

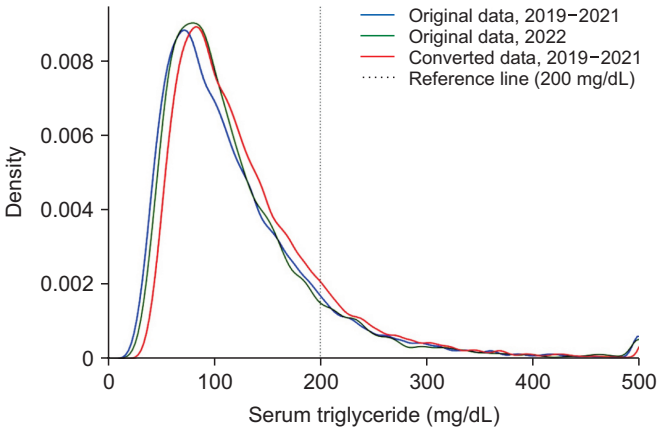


Fig. 3. Serum triglyceride levels in KNHANES participants before and after conversion. Kernel density plots represent serum triglyceride data from KNHANES 2019–2021 (N=16,015) and KNHANES 2022 (N=4,996), excluding pregnant women and non-fasting individuals (<12 hrs), before and after adjustment using the conversion equation y (glycerol non-blanking, mg/dL)=11.94 (mg/dL)+0.99x (glycerol blanking). The blue curve represents data before conversion, and the red curve shows data after conversion. The X-axis represents serum triglyceride levels, and the Y-axis depicts the fraction of the population in density plots, estimated using kernel density functions with the area under the curve integrated into one. The vertical gray dotted line marks the triglyceride level of 200 mg/dL (2.26 mmol/L), as defined by the National Cholesterol Education Program Adult Treatment Panel III criteria for high triglyceride levels. For presentation purposes, triglyceride values ≥500 mg/dL (5.65 mmol/L) were converted to 500 mg/dL (5.65 mmol/L).
Abbreviation: KNHANES, Korea National Health and Nutrition Examination Survey.

the conversion equation *y* (mg/dL) = 11.94 (mg/dL)+0.99*x* to data from 2019–2021, the triglyceride concentrations shifted to the right (red line), resulting in an increased prevalence of hypertriglyceridemia (≥200 mg/dL [2.26 mmol/L]) from 11.9% to 13.8% (unweighted) during 2019–2021.

During the KNHANES (2018–2022), each laboratory participated in CDC LSP surveys, and the results were available for triglyceride measurements. Fig. 4 displays the percent bias estimation for each round of CDC LSP survey materials from the KNHANES from 2018 to 2022. As the target values for triglycerides in the CDC LSP were assigned to total glycerides, the KNHANES triglyceride data for 2018–2021 (which eliminated free glycerol from total glycerides using the glycerol blanking method) differed significantly from the target values of the CDC LSP (exceeding −5% in terms of the bias limit, within −15% for the total allowable error limit). Conversely, the glycerol non-blanking methods used in the KNHANES during 2022 showed acceptable results, with biases within ±5.0% according to NCEP ATP III criteria for bias estimations.

DISCUSSION

The historical evolution of triglyceride measurement methods and assay standardization reflects advancements in technology, improved scientific understanding, and the pursuit of accuracy and reliability [3]. Standardizing analytical methods, including proficiency testing with reference materials and expanding global clinical research, has ensured consistent and comparable triglyceride measurements across diverse populations and various settings [3, 6, 16, 21–24].

We developed a conversion equation, *y* (triglyceride concentration without glycerol blanking; mg/dL) = 11.94 (mg/dL)+0.99*x*

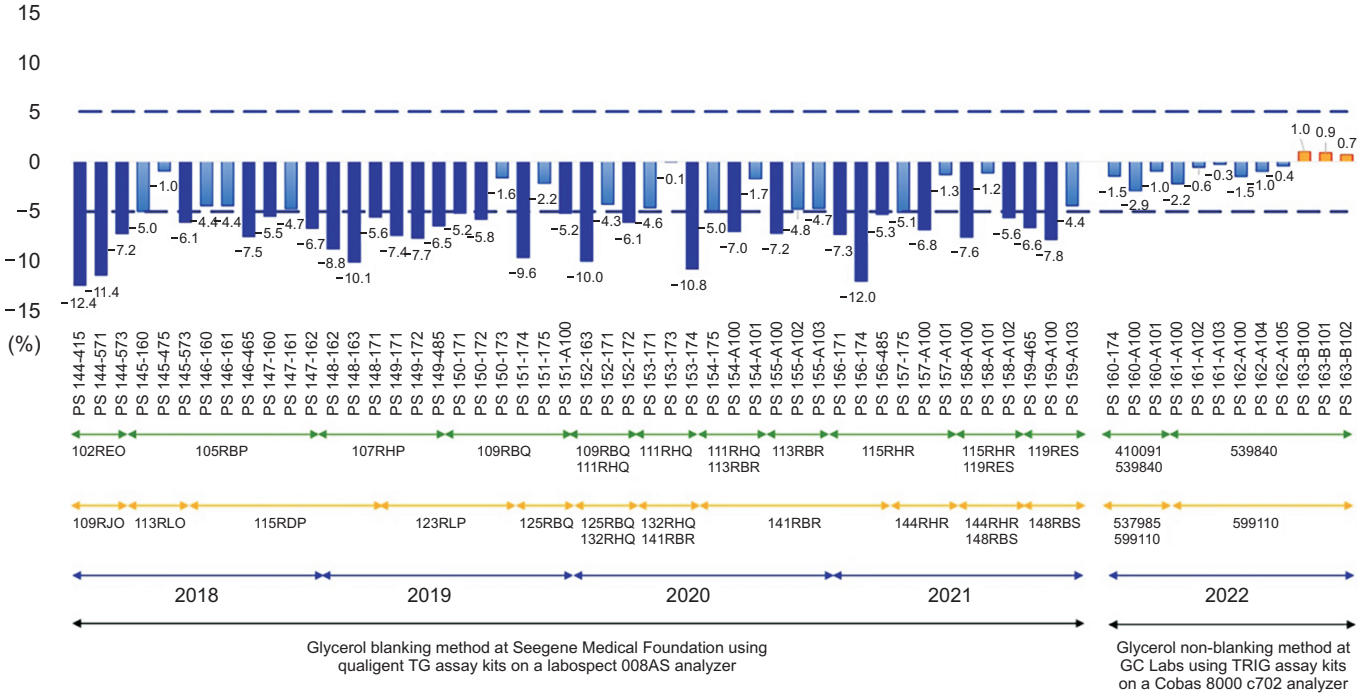


Fig. 4. Percent bias change for triglyceride measurements from the CDC Lipid Standardization Program (LSP) during KNHANES (2018–2022). During KNHANES 2018–2021, triglyceride measurements were performed using glycerol blanking methods at Seegene Medical Foundation, whereas in KNHANES 2022, measurements were performed using glycerol non-blanking methods at GC Labs. The X-axis represents each survey round and specimen identifier (labeled as PS-survey round-specimen identifier). Bars represent percent bias estimates calculated by the CDC LSP for each round. The blue dotted lines indicate the upper and lower allowable bias limits (±5%) defined by NCEP ATP III criteria. Green arrows denote calibrator lots used during survey rounds; yellow arrows indicate reagent lots; blue arrows represent the survey year; and black arrows show the analytical methods used during these rounds. The CDC LSP survey sets target values using glycerol non-blanking methods, represented by a thin gray line at 0.0%. Blue bars represent results below the target values assigned by CDC mean values, orange bars indicate results above target values, and dark blue bars represent results exceeding the ±5% bias limit. Percent bias was calculated using the formula: (Result from each lab − CDC mean [target])/CDC mean (target)×100.
Abbreviation: KNHANES, Korea National Health and Nutrition Examination Survey.

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(triglyceride concentration with glycerol blanking; mg/dL), which demonstrated a strong correlation (r=0.997) and an average bias of 10.0%. In mmol/L, the equation was $y=0.13$ (mmol/L)+$0.99x$. This finding aligns with previous data showing that y (triglyceride concentration without glycerol blanking; mmol/L)=0.1623 (mmol/L)+$1.012x$ (triglyceride with glycerol blanking; mmol/L), with a conversion factor of mmol/L×88.5=mg/dL (corresponding intercept y: 14.36 mg/dL) [3]. Prior studies have reported an average bias of 16.7% higher triglyceride concentrations with glycerol non-blanking methods than with glycerol blanking methods, with an estimated bias of 8.2% at a triglyceride concentration of 203 mg/dL (2.29 mmol/L) [3]. These results are consistent with findings from the KAEQAS lipid proficiency testing program (2016–2018) [25], which indicated a total glyceride concentration (glycerol non-blanking, mg/dL) of 6.87 (mg/dL)+1.01×triglyceride concentration with glycerol blanking (mg/dL) for target-value comparison [total glyceride (mmol/L)=0.08 mmol/L+1.01×triglyceride with glycerol blanking (mmol/L)]. When comparing the average values of all participants, the proficiency testing material showed a total glyceride concentration (glycerol non-blanking, mg/dL) of 8.04 (mg/dL)+0.99× the triglyceride concentration with glycerol blanking (mg/dL) [total glycerides (mmol/L)=0.09 mmol/L+0.99×the triglyceride with glycerol blanking (mmol/L)] [25]. Our current findings confirm these observations, indicating a generally greater negative bias in specimens with lower triglyceride concentrations [7, 9, 14, 26]. Free glycerol levels can increase due to factors such as exercise, liver disease, diabetes, hemodialysis, stress, and glycerol-containing medications, whereas decreases may occur in lipid metabolism disorders or treatments affecting lipolysis. Substances such as vitamin C, acetaminophen, N-acetyl cysteine, and bilirubin can cause negative bias in glycerol non-blanking methods, and variability in sample handling can further contribute to inaccuracies [2, 3, 9, 14, 26, 27]. Caution is required when applying the conversion equation in individuals with altered free glycerol concentrations. Future studies should evaluate the impact of analytical method changes on these subgroups [9, 27]. However, quantifying free glycerol to determine true triglyceride concentration is typically of limited importance in most clinical scenarios, as the potential error due to free glycerol is <9 mg/dL (0.10 mmol/L) in 99% of individuals [9].

Standardization of analytical methods is essential to ensure accuracy and consistency with reference measurement procedures (RMPs) for analytes across various measurement systems, laboratories, and time frames [22–24]. The primary RMP and standard reference material (SRM), namely the National Institute of Standards and Technology (NIST) SRM 1595 Trialmitin (a pure chemical SRM certified in 1983 and updated to NIST SRM 1595a in March 2024 for measuring triglycerides) were established by the NIST [28]. The US CDC developed the secondary RMP and secondary SRM in 1963, initially using chemical methods to reduce free glycerol, which were cumbersome for routine clinical applications [7, 9, 14]. These methods were replaced by isotope dilution mass spectrometry (IDMS) in 2012 to measure total glycerides, thereby improving standardization of routine clinical serum-glyceride measurements considering that most clinical laboratories now use glycerol non-blanking methods [9, 14]. During RMP development, NIST SRM 1595 was used as a calibrator, and NIST SRM 1951b (comprising a frozen human serum matrix with certified concentrations of total glycerides [I and II] and triglycerides [I]), along with additional frozen serum samples, were utilized for precision and bias estimations. However, a direct comparison between older enzymatic chromotropic acid methods and the newly developed CDC IDMS method was not performed [7].

The National Cholesterol Education Program (NCEP) recommends the CDC reference method as the standard, and NCEP ATP III-derived guidelines were based on CDC-developed secondary RMPs [5, 7, 9]. CDC LSP certification is valuable for estimating bias in terms of target concentrations and precision [7, 15, 29]. Using CDC LSP is crucial for comparing epidemiological data, as demonstrated by Finland's National Public Health Institute, which ensures accuracy through robust quality-assurance methods [14, 15, 23, 24, 29, 30]. In Korea, the National Health Insurance Service (NHIS) generates triglyceride data through its health check-up program for most residents, with data generated by hundreds of hospitals using diverse analytical methods [31]. According to a 2016–2018 KAEQAS report, two-thirds of laboratories participating in the accuracy-based lipid proficiency testing program used glycerol non-blanking methods [23, 25]. During the second survey performed in 2022, 76.4% of survey participants used glycerol non-blanking methods for triglyceride measurements (approximately half of them used analytical methods by Roche) [25]. This proportion is comparable with data from CDC LSP-certified participants [29].

We evaluated the impact of changing analytical methods on triglyceride measurements within the KNHANES framework. Even minor systematic biases may have significant clinical and epidemiological implications. However, discrepancies in test values that do not affect classification thresholds may not alter prevalence rates [9, 14]. Considering that most clinical laboratories globally use glycerol non-blanking methods and that clini-

cians interpret triglyceride results without detailed knowledge of the analytical methods used, researchers relying on KNHANES data must recognize these changes. Therefore, applying the conversion equation judiciously is essential to preserve research integrity [5].

Our study has limitations, including differences in population characteristics, an uneven distribution of triglyceride concentrations, and a lack of data on reagent and calibrator lots. Comparisons were restricted to a single lot of reagents and calibrators for each analytical method, which may not account for potential variations between different lots employed. Variations in sample sizes between KNHANES data and the conversion equation study, which used a relatively small sample size, might have influenced the observed differences and could have affected the generalizability of our findings, raising concerns about potential overcorrections when applying the conversion equation. Further studies must be conducted to evaluate population-specific prevalences, demographic differences, and their underlying causes and impacts, as statistical correction methods can vary depending on the specific research objectives and population characteristics. Furthermore, cross-validation must also be performed to assess the impact of transitioning from glycerol blanking to non-blanking methods on cardiovascular disease risk monitoring in epidemiological studies.

In conclusion, we developed conversion equations to align historical data with current triglyceride measurement methods, ensuring robust and valid conclusions in longitudinal studies. Our findings contribute to the methodological literature on triglyceride measurements and offer a model for other epidemiological surveys undergoing similar transitions. Global standardization of biochemical triglyceride measurements in health surveys improves comparability and provides critical insights for guiding future methodological decisions in similar health examination surveys worldwide.

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AUTHOR CONTRIBUTIONS

Choi R and Seo JD: conceptualization, data curation, formal analysis, methodology, resources, writing – original draft, writing – review & editing; Cho EJ: data curation, formal analysis, writing – review & editing; Lee W: methodology, resources, validation, funding acquisition, writing – review & editing; Yun YM: conceptualization, formal analysis, methodology, project administration, resources, supervision, writing – review & editing. All authors reviewed and approved the manuscript.

CONFLICTS OF INTEREST

None declared.

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